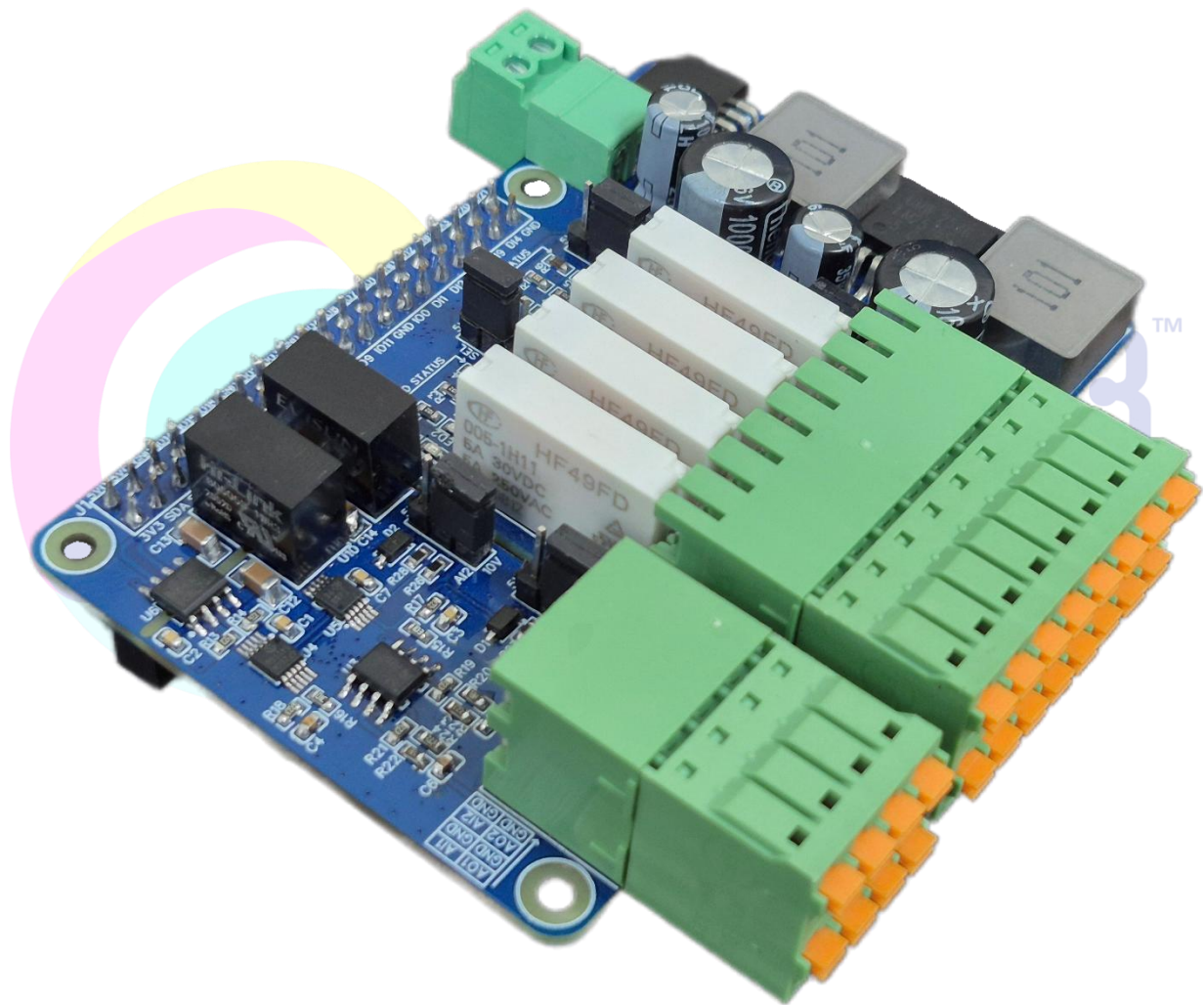


Technical Datasheet of WIN-PiHAT-4DD-2AIAO

1. Overview

The **PiHAT 4DD-2AIAO** is a versatile mixed-signal expansion board for Raspberry Pi 4 and CM4, offering **comprehensive I/O capabilities for industrial automation and control**. It provides precise signal measurement and robust load switching by combining **4 opto-isolated Digital Inputs, 4 relay-driven Digital Outputs, 2 high-resolution Analog Inputs, and 2 Analog Outputs**. This ensures seamless data collection and actuation between external field devices and the Pi's GPIO header for reliable operation in industrial environments.



This PiHAT features a **scalable architecture** supporting **independent analog and digital channels**. Its **consolidated mixed-signal design** allows maximum I/O flexibility while maintaining a **compact and reliable solution** for embedded sensing, load automation, and precise industrial control applications.

Specifications

Category	8AI PiHAT
Input Type	Digital: Opto-isolated Digital Inputs Analog: 12-bit ADC Resolution (Selectable 0-5V, 0-10V, 4-20mA)
Output Type	Digital: Relay-Isolated Digital Outputs (5A @ 250VAC/30VDC) Analog: 12-bit DAC Resolution (0-10V Range)
Channel Count	12 Channels Total (4 DI, 4 DO, 2 AI, 2 AO)
Connectors	Terminal Blocks (3.81 mm Pluggable): • One 2x8-Pin (for all DI & DO channels) • One 2x4-Pin (for all AI & AO channels) • One 2-Pin (24V DC; powers PiHAT & Raspberry Pi simultaneously)

General :-

Dimensions	56 mm L x 85 mm W
Power	Input Power – 24V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



AI (Analog Inputs) :-

Channels	2
ADC Resolution	12-bit
Input Type	Single-Ended / Differential (Depending on config)
Voltage Input Range	0 ~ 5V DC, 0 ~ 10V DC
Current Input Range	0 ~ 20mA, 4 ~ 20mA
Input Impedance(V)	≥ 1 MΩ
Input Impedance(I)	~ 250 Ω (Typical for mA drop)
Accuracy	±0.1% to ±0.3% FSR (Full Scale Range)
Sample Rate	Programmable (e.g., 128 SPS to 3300 SPS)

AO (Analog Outputs) :-

Channels	2
DAC Resolution	12-bit
Output Type	Single-Ended Voltage Output
Voltage Output Range	0 ~ 10V DC
Max Output Current	35 mA (Short-Circuit limit), ~15 mA continuous
Accuracy	±0.1% to ±0.3% FSR (Full Scale Range)
Settling Time	6 µs (Typical)

DI (Digital Inputs) :-

Channels	4
Input Type	12V & 5V with Reference, 80V Absolute Max. @ 50mA
Sense Voltage	3.3 – 12V DC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

DO (Digital Outputs) :-

Channels	4
Output Type	Relay-Isolated Digital Outputs (5V SPST-NO Industrial Relay)
Contact Form	1A (SPST-NO)
Contact Material	AgNi Alloy
Contact Capacity	5A @ 250VAC, 5A @ 30VDC
Coil Voltage	5VDC
Coil Power	0.12W
Insulation Resistance	≥ 100MΩ
Electrical Life	1x10 ⁵ operations
Mechanical Life	1x10 ⁷ operations
Operating Time	≤10 ms (max., at rated voltage)
Release Time	≤05 ms (max., at rated voltage)

Configuration :-

The PiHAT 4DD-2AIAO operates as a comprehensive mixed-signal expansion board, interfacing 4 Digital Inputs, 4 Digital Outputs, 2 Analog Inputs, and 2 Analog Outputs to the Raspberry Pi GPIO header. Data acquisition and load control are managed simultaneously via software running on the Raspberry Pi.

Analog Configuration (AI & AO): The 2 AI and 2 AO channels are managed over a galvanically isolated I2C communication bus utilizing the ISO1540DR isolator. When an external analog signal is applied to an input terminal, the onboard ADS1015 ADC converts it into a precise 12-bit digital value. Conversely, software-generated 12-bit digital commands are sent to the onboard MCP4728 DAC, which immediately converts them into continuous, proportional analog voltages (0-10V) to drive external equipment.

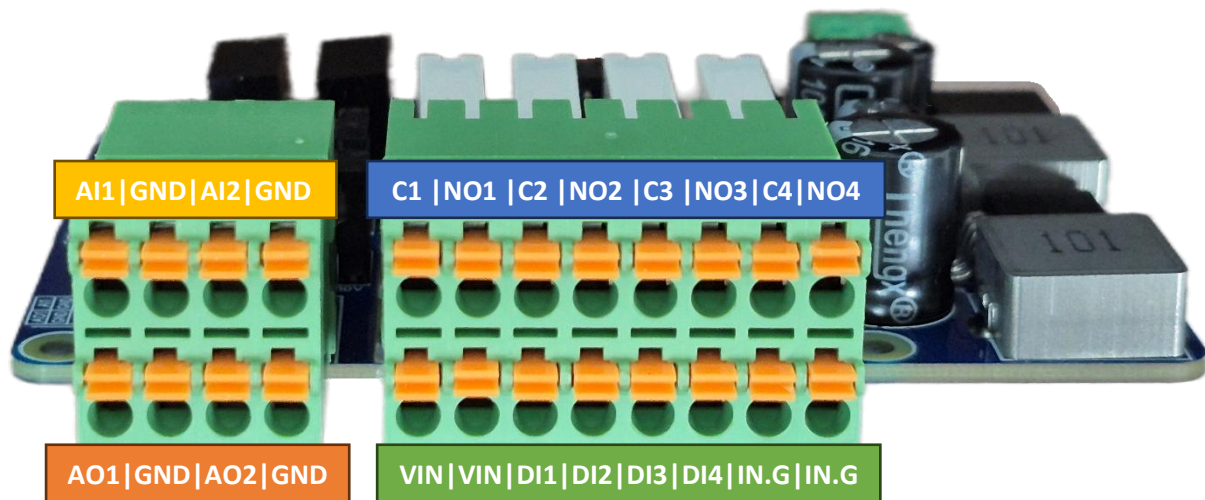
Digital Configuration (DI & DO): The digital channels bypass the I2C bus and interface directly with dedicated Raspberry Pi GPIO pins. The 4 DI channels use optocouplers to safely step-down external signals, allowing your software to read active states as standard logic levels. The 4 DO channels are controlled by driving their assigned GPIO pins HIGH to energize the onboard relays (closing the NO contact) or LOW to de-energize them.

The following are step by step guides for configuration:

1. Connect the Board to the Raspberry Pi via header pins
2. Connect the Module to 24V DC POWER SUPPLY.

I/O	Signal Name	Pi GPIO	Header Pin Number	Description
2AI & 2AO	SDA	02	03	Isolated I2C data line for ADS1015 (AI) & MCP4728 (AO)
	SCL	03	05	
4DI	DI1	05	29	Direct GPIO monitoring for opto-isolated inputs
	DI2	06	31	
	DI3	16	36	
	DI4	26	37	
4DO	DO1	22	15	Direct GPIO control for relay-driven outputs
	DO2	23	16	
	DO3	24	18	
	DO4	25	22	

Wiring Diagram & Terminal Configuration:-



Analog Inputs (AI1 – AI2)

- **Terminals:** Connect the sensor's positive signal wire to the **AIx** terminal and the common/ground wire to the corresponding **GND** terminal.
- **Configuration:** Each AI channel features a dedicated hardware jumper. Place the jumper on the **5V** position to read 0-5V signals, or the **10V** position for 0-10V signals. Always ensure this setting strictly matches your sensor's maximum output to prevent data clipping.



Analog Outputs (AO1 – AO2)

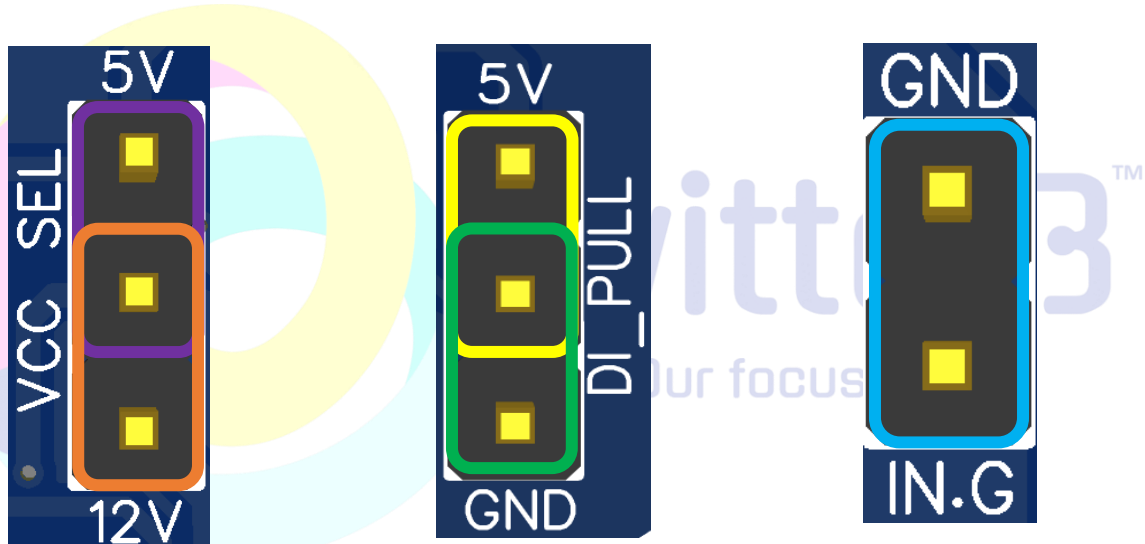
- **Terminals:** Connect the receiving device's signal input to the **AOx** terminal and its ground reference to the **GND** terminal. The DAC provides a proportional 0-10V control signal across these pins.

Digital Outputs (DO1 – DO4)

- **Terminals:** Each relay channel utilizes a **Cx** (Common) and **NOx** (Normally Open) terminal pair.
- **Wiring:** For load switching, connect the load's power supply (Phase/Positive) to **Cx**, and connect the load itself to **NOx**. When the software drives the GPIO high, the relay energizes, closing the internal circuit between Cx and NOx.

Digital Inputs (DI1 – DI4)

- **Terminals:** Connect the switching signals to the **DIx** terminals. Depending on your switching logic, DI signals are measured with respect to either **VIN** (Voltage Input Reference) or **IN.G** (Input Ground).
- **Configuration:** Use the onboard **DI_PULL** jumper to set the pull state (High/Low) for the optocouplers.
- **External Power Warning:** If your application uses an external reference voltage source to drive the digital inputs, you must completely remove the associated internal jumpers (**VCC_SEL** and **IN.G**) to ensure safe electrical isolation.



Sample Code & Examples :-

Access the official Python example scripts for **4DI, 4DO, 2AI, and 2AO control and testing** from our repository here:

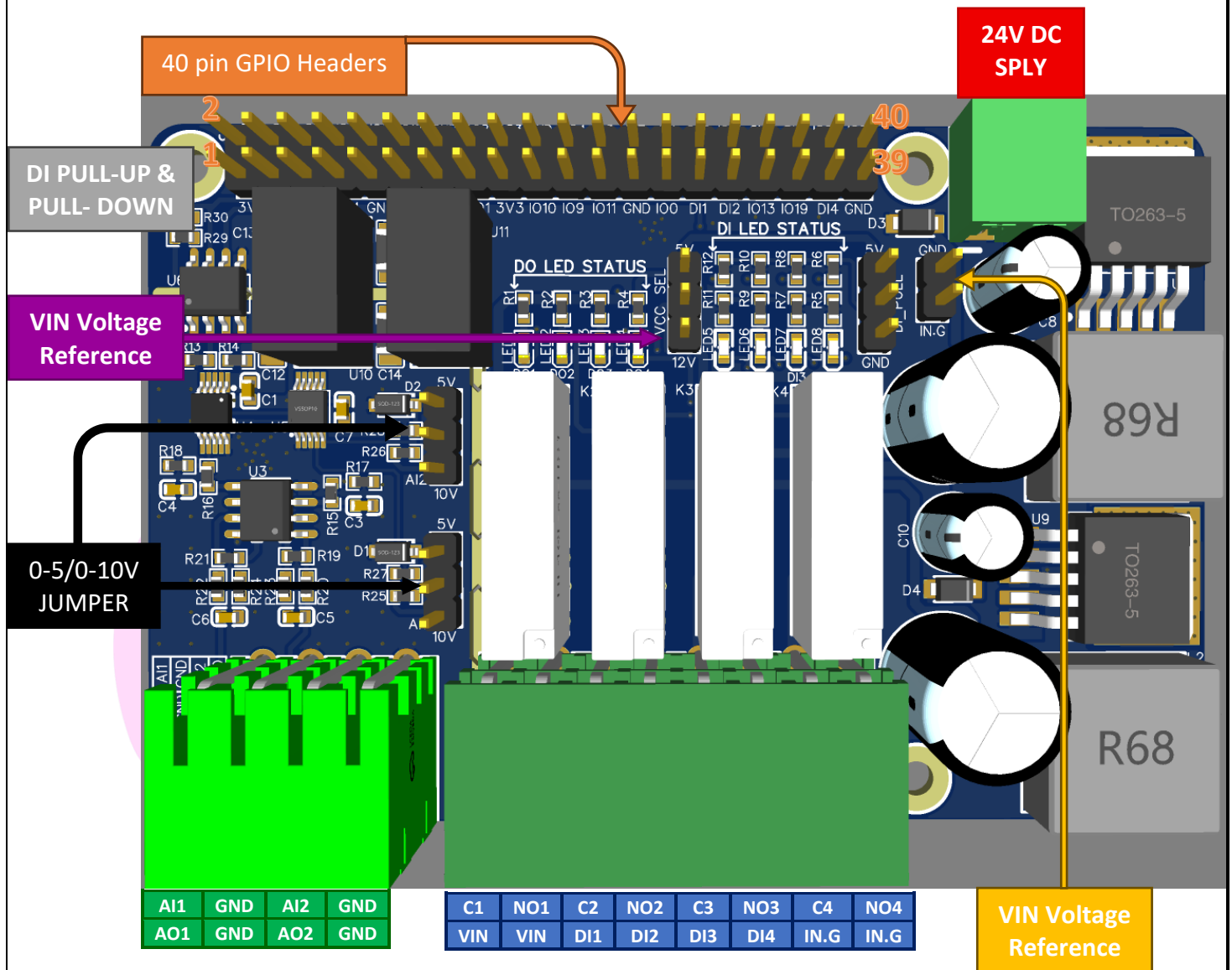
- [WIN-PIHAT-4DD-2AIAO](#)



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Our focus is You

WIN-PIHAT-4DD-2AIAO

Hardware Interface & Connector Layout:



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